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## Technical Memorandum — Geotechnical Assessment of Pier P3 Founding Conditions

Nandira River Bridge Project — Package NRB-4

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| <b>Document Reference:</b> | TM-NRB4-009                                                                |
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| <b>Contract No.:</b>       | NHIA/NRB4/CW/2020-01                                                       |
| <b>From:</b>               | Meridian Engineering Consultants, Geotechnical Section                     |
| <b>To:</b>                 | Project File                                                               |
| <b>CC:</b>                 | National Highways Infrastructure Authority; Sagara Constructions Pvt. Ltd. |

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### Subject: Geotechnical Assessment — Pier P3 Founding Conditions, North-East Quadrant

1. **BACKGROUND:** Further to Notice CTR-NRB4-0061 and Daily Progress Report DPR-0920-2021, this memorandum records the Engineer's assessment of ground conditions in the north-east quadrant of the Pier P3 founding level, following trial pitting carried out between 24-Sep-2021 and 30-Sep-2021.
2. **FINDINGS:** Trial pits TP-1 to TP-3 confirm loosely compacted backfill to depths ranging from 2.7 to 3.2 metres below design founding level, occupying an area of approximately 4 metres by 4 metres within the north-east quadrant. This is consistent with backfilling of the excavation left by removal of the 132kV tower foundation referred to in VPGC-NRB4-0045. The backfill material does not meet the bearing criteria stated in the Specification and referred to in Employer's Requirements Section 6, and is not suitable for direct founding of the pile cap as designed.
3. **ASSESSMENT:** Two options were considered: (a) full removal of the backfill and replacement with mass concrete fill to founding level, or (b) local enlargement and thickening of the pile cap with additional reinforcement to redistribute load away from the affected zone, combined with partial mass concrete replacement of the weakest backfill material. Option (b) is recommended as it avoids the programme risk of a substantially deeper excavation immediately adjacent to the completed Pier P3 shaft reinforcement cage layout, and is compatible with the founding levels already achieved elsewhere across the pile cap.
4. **RECOMMENDATION:** The Engineer will issue a revised drawing for the Pier P3 pile cap incorporating local enlargement, additional thickness, partial mass concrete replacement, and additional reinforcement in the affected zone, and will instruct this as a Variation under Sub-Clause 13.1, the condition not being one addressed by the original design.

Prepared by: Geotechnical Section, Meridian Engineering Consultants